vessel resulting in smaller work done.
4. The gas is not ideal. So the collision is not elastics.
5. The wire conducts some of the heat energy from the heater.

2 (b)(ii)1.

In the first experiment, **the piston is fixed and so no work is done.** All the 150 J goes into increasing the internal energy of the gas. [B1]

In the second experiment, the piston is free to move and so work is done by the gas. Because there is **work done by the gas, there is less thermal energy** available to increase the internal energy and so **the temperature rise for the second experiment is less**. [B1]

Using 1st law of thermodynamics

Expt 1 : $\Delta U = Q + 0$

Expt 2 : $\Delta U' = Q - W$ (Q is 150 J for both)

Thus, the rise in internal energy (and temperature) for Expt 2 is less.

2 (b)(ii)2.

$\Delta U = Q + W$

$= 150 + 0$

$= 150$ J [A1]

2 (b)(ii)3.

efficiency = Work / heat energy supplied

$40/100 = W / 150$ [C1]

$W = 60$ J [A1]

Deduct one mark for negative answer

2 (b)(ii)4.

$\Delta U = Q + W$

$= 150 + (-60)$

$= 90$ J [A1]

Allow ECF

3 (a)

The principle of superposition states that when two or more waves meet at a point [B1], the resultant displacement at that point is equal to the vector sum of the displacements of the individual waves at that point. [B1]

3 (b)(i)

It means that the signals from P and Q reaching the point are in antiphase. [A1] (accept completely or 180° or π out of phase.)

3 (b)(ii)

Along the line joining X and Y, any point Z along XY will be such that the distance PZ equals the distance QZ. Thus the path difference PZ – QZ will be zero. [B1]

The phase difference of the two signals reaching point Z will be due to the phase difference between the transmitter P and Q which is π out of phase.

Thus the waves from P and Q meet point Z in antiphase and the resultant displacement is **always zero** [B1] and the signal detected by the ship is zero.

3 (b)(iii)

λ wavelength of wave = v / f = (3.0 x 10^8 m s^{-1}) / (1.0 x 10^6 Hz) = 300 m

[M1]

When the ship detects an amplitude that is twice, it reaches a point of constructive interference. [M1]

Distance between destructive interference to constructive interference = $\lambda/4$

= 75 m [A1]

3 (b)(iv)

Distance from mid point to Q = 30 km.

Distance between two nodes = $\lambda/2$ = 150 m

Number of dips = 30000/ 150 = 200

4 (a)(i)

As the graph shows a significant drop in the p.d across the variable resistance as current changes, it shows that the variation of p.d in internal resistance is significant

4 (a)(ii)

Using V = E - Ir

Thus, when I = 0, V = E [B1]

When extending the graph, the line cut 4.4 V, E = 4.4 V

Gradient of graph = -r

Thus -r= (3.1-0.7)/(1.2-3.4) = -1.1 [B1]

r = 1.1

4 (b)

When device is operating in normal condition,

Resistance of device = 0.53Ω [B1]

Effective resistance of circuit needed = 4.4/1.5 = 2.93 [B1]

Additional resistance required = 2.93 – 0.53 – 1.1 = 1.3 Ω

[Draw an additional 1.3 Ω of resistor in series with the emf and device] [A1]

5 (a)

Number of electrons removed = total charge/ charge of 1 electron

$= \dfrac{1.11\times10^{-6}}{1.6\times10^{-19}}$ [C1]

$= 6.94 \times 10^{12}$ [A1]

5 (b)

(a) *Values for graph*

Distance/ m	E/ $\times 10^3$ V
0.50	40.0
1.00	10.0
1.50	4.44
2.00	2.50

Correct points [B1]

Shape for curve with decreasing magnitude with increasing distance. [B1]

5 (c)(i)

Work done = qΔV = $\dfrac{q^2}{4\pi\varepsilon_0}\left(\dfrac{1}{1.2}-0\right)$

=$(1.11 \times 10^{-6})^2(9.0 \times 10^9)(0.833)$ [M1]
= 9.24×10^{-3} J [A1]

<u>*5 (c)(ii)*</u>

Field strength due to charge of right = $\dfrac{q}{4\pi\varepsilon_0(0.5)^2}$ *points to the left.*

Field strength due to charge of left = $\dfrac{q}{4\pi\varepsilon_0(0.7)^2}$ *points to the right.*

Resultant field strength = $\dfrac{q}{4\pi\varepsilon_0(0.5)^2}-\dfrac{q}{4\pi\varepsilon_0(0.7)^2}$ *[C1]*

= 1.96×10^4 V m^{-1}[A1]

Direction towards the left [A1] (accept negative sign if student states rightward as convention.

<u>*6 (a)*</u>